

Mathematics, 2015 EC, entrance exam, version 2

Grade 12 · Mathematics · 2015 E.C.

Time allowed: 120 minutes

Questions: 51

Total marks: 51

Name _____ Grade _____ ID No. _____ Date _____

1. If $f(x) = \frac{\sqrt{x+2}}{x+2}$ and $g(x) = \frac{1}{x} - 2$ then what is the value of $f(g(x))$ for all $x > 0$? (1)
- A. $\sqrt{x} - 2$ B. $\sqrt{x} + 2$
C. $\sqrt{x}$ D. $1/\sqrt{x}$
-
2. Which of the following is Not the property of $f(x) = x^{2/5}$? (1)
- A. $f(xy) = f(x)f(y)$
B. decreasing on $(-\infty, 0]$
C. the range is real number
D. the graph is symmetric with respect to the y-axis
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3. What is the partial fraction decomposition of $\frac{5x^2 - 8x + 5}{(x-2)(x^2 - x + 1)}$? (1)
- A. $2/(x-2) + (3x-1)/(x^2 - x + 1)$ B. $3/(x-2) + (2x-1)/(x^2 - x + 1)$
C. $1/(x-2) + (4x-3)/(x^2 - x + 1)$ D. $4/(x-2) + (x+1)/(x^2 - x + 1)$
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4. Suppose $f(x) = \frac{Q(x)}{x(x^2-1)}$ where $Q(x)$ is a quadratic function. Which one of the following is necessarily true about the graph f ? (1)
- A. $x=0$, $x=1$ and $x=-1$ are vertical asymptote of the graph f
B. The graph of f does not intersect with its horizontal asymptote.
C. The vertical asymptote of the graph of f is only $x=-1$, if $Q(x)=x^2-x$
D. The vertical asymptote of the graph of f is only $x=1$, if $Q(x)=2x^2$
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5. Given line $4y = 3x + 24$ and another line ℓ parallel to this line at a distance of 4 units from it. A possible equation for ℓ is (1)
- A. $y = (3/4)x$ B. $y = (3/4)x + 1$
C. $y = (3/4)x + 6$ D. $y = (3/4)x - 6$
-
6. Suppose an eccentricity of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is reciprocal to that of the eccentricity of the ellipse $x^2 + 4y^2 = 4$ if the hyperbola passes through a focus of the ellipse, what is the equation of the hyperbola? (1)
- A. $x^2 - 2y^2 = 2$ B. $x^2 - 3y^2 = 3$
C. $x^2 - y^2 = 1$ D. $2x^2 - y^2 = 2$
-
7. Which of the following is NOT true about a parabola $y^2 + 2x - 4y + 6 = 0$? (1)
- A. It opens to the left B. Its vertex is $(-1, 2)$
C. Its focus is $(3/2, 2)$ D. Its directrix is the line $x = -1/2$
-
8. The distance from the point $P(3,4)$ to the line L with equation: $3x + 4y = 5$ is (1)
- A. 6 units B. 7 units
C. 4 units D. 5 units

9. What is the locus of all points in the plane in which the difference of the distances from two fixed points is constant? (1)
- A. Ellipse
B. Circle
C. Hyperbola
D. Parabola
-
10. Let p and q be propositions. Which of the following compound statement is a tautology? (1)
- A. $(p \wedge q) \Leftrightarrow p$
B. $(p \Rightarrow q) \Leftrightarrow (\neg p \vee q)$
C. $(p \vee q) \Leftrightarrow p$
D. $(p \Rightarrow q) \Leftrightarrow q$
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11. If $p \Rightarrow q$ is false, then which of the following compound statement is true? (1)
- A. $p \wedge q$
B. $p \Leftrightarrow q$
C. $p \vee q$
D. $(p \vee q) \Rightarrow q$
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12. Which of the following arguments is invalid logical argument? (1)
- A. $p \wedge \neg q, p, r \Rightarrow q \vdash \neg r$
B. $p \vee r, p \wedge q \vdash p \Rightarrow \neg q$
C. $p \Rightarrow q, \neg r \Rightarrow \neg q \vdash \neg p \vee r$
D. $\neg p \vee q, \neg q \vdash \neg q$
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13. Which of the following is the inverse of; "If wages increase, then inflation decreases"? (1)
- A. If inflation decreases, then wage increase.
B. If wages decreases, then inflation increases.
C. If wages decreases, then inflation decreases.
D. If wages increase, then inflation increases.
-
14. What is the odds in favor of getting exactly two heads when three coins are tossed? (1)
- A. 3/8
B. 5/8
C. 5/3
D. 3/5
-
15. From a box containing 2 white and 6 green balls, a ball is drawn at random. The probability of not a green ball is: (1)
- A. 1/4
B. 3/4
C. 1/3
D. 2/3
-
16. How many three digit number can be formed from the digits 2,5,6,7 and 9 less than 600, if digits is used once only? (1)
- A. 24
B. 40
C. 125
D. 50
-
17. In a random arrangement of the word "SOCIETY", what is the total number of words in which all the three vowels come together? (1)
- A. 120
B. 720
C. 420
D. 840
-
18. Which of the following is the constant term of $(x^2 - 1/(2x^6))^8$? (1)
- A. 28
B. 7
C. -7
D. -28

19. From the above data (marks distribution), if students in the top 8% are to be awarded a certificate, what is the minimum mark for a certificate? (Data: class 10-14 (15), 15-19 (23), 20-29 (46), 30-39 (16)) (1)
- A. 34.5
B. 30.5
C. 32
D. 33
-
20. A group of 100 items have a mean of 62. If the mean of 40 of these items is 55, what is the mean of the remaining items? (1)
- A. 66.67
B. 69.00
C. 65.00
D. 64.55
-
21. If $[[1-x, 0], [1, 2y]] + [[y-1, 4], [2, 1-3x]] = [[0, 4], [3, 5]]$, then x and y respectively? (1)
- A. -2 and 2
B. 1 and -1
C. 0 and 0
D. -4 and -4
-
22. If A is a square matrix of order 3 and $\det(A)=5$, then what is the value of $\det(A \operatorname{adj}(A))$? (1)
- A. 125
B. 25
C. 5
D. 3
-
23. For any $n \times n$ square matrix A, which of the following is true? (1)
- A. $\det(A) = -\det(A^T)$
B. If k is scalar, $\det(kA) = k^n \det(A)$
C. If B is obtained from A by interchanging two rows, then $\det(A)=\det(B)$
D. If A is invertible, then $\det(A)=\det(A^{-1})$
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24. For what value of x does $((1+i)/(1-i))^x = 1$? (1)
- A. $x=2n+1$
B. $x=2n$
C. $x=4n$
D. $x=4n+1$
-
25. If the real part of $(z+2)/(z-1)$ is 4, then the locus of z is: (1)
- A. Circle
B. Straight line
C. Parabola
D. Hyperbola
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26. If $z = (2-2\sqrt{3}i)/(1+i)$, then the polar representation is: (1)
- A. $2\sqrt{2}(\cos(-105^\circ)+i \sin(-105^\circ))$
B. $4(\cos(-105^\circ)+i \sin(-105^\circ))$
C. $2\sqrt{2}(\cos 15^\circ+i \sin 15^\circ)$
D. $4(\cos 75^\circ+i \sin 75^\circ)$
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27. If $z_1 = (2-i)/(1+i)$ and $z_2 = (1+i)/(1-i)$, then $z_1 - 2z_2 = ?$ (1)
- A. $1-7i$
B. $(1-7i)/2$
C. $(1+7i)/2$
D. $1-7i$
-
28. The 15th term of the sequence {0, -3, -8, -15, ...} is (1)
- A. -245
B. -255
C. -224
D. -242
-
29. The sum of the first four terms of an AP is 56 and sum of last four terms is 112. First term 11, find number of terms. (1)
- A. 10
B. 18
C. 12
D. 11

30. A ball is dropped from 10m, rebounds 50% each time. Total vertical distance? (1)
- A. 20m B. 40m
C. 45m D. 30m
-
31. Sum of series $\sum_{n=1}^{30} (-1)^n (1/n + 1/(n+1))$ is: (1)
- A. -30/31 B. 29/30
C. -29/30 D. 30/31
-
32. Third term of GP is 4/9, sum to infinity is 3. Find first term. (1)
- A. 1/2 B. 2/3
C. 1/3 D. 1
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33. $\lim_{n \rightarrow \infty} (1^2 + 2^2 + \dots + n^2) / ((n^2 - 1)(n + 1)) = ?$ (1)
- A. 0 B. 1/2
C. 1/3 D. does not exist
-
34. $\lim_{x \rightarrow 1/2} (2x - 1) / (1 - \sqrt[3]{2x}) = ?$ (1)
- A. 1 B. 3
C. -3 D. -1.5
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35. Which statement is NOT true? (1)
- A. A null sequence has limit 0
B. Increasing bounded above sequence converges to lub
C. Decreasing bounded below sequence converges to glb
D. Every bounded sequence is convergent
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36. Let $f(x) = \{ a \sin x / (x - |x|) \text{ for } x < 0, e^{-x} \text{ for } x \geq 0 \}$. If f continuous at 0, find a . (1)
- A. 4 B. -4
C. 1/2 D. 2
-
37. $\lim_{x \rightarrow \infty} (\sqrt{x^2 + 2x} - x) = ?$ (1)
- A. 0 B. 1
C. ∞ D. does not exist
-
38. $\lim_{x \rightarrow 0} (2x + \tan x) / (x \sec x) = ?$ (1)
- A. 2 B. 0
C. 1 D. 3
-
39. $\lim_{x \rightarrow \infty} ((x+1)/x)^{2x+4} = ?$ (1)
- A. e B. e^2
C. e^4 D. ∞
-
40. If $f(x) = x^2 + 2 \ln x^2$, then $\lim_{h \rightarrow 0} (f(1+h) - f(1)) / h = ?$ (1)
- A. 2 B. 6
C. 5 D. 0

41. If $f(x)=\pi^3$, then $f'(2)=?$ (1)
- A. $3\pi^2$ B. 0
C. 12 D. 1
-
42. If $y=e^{2x}+x-3\cos x$, then $d^2y/dx^2 = ?$ (1)
- A. $4e^{2x}+3\cos x$ B. $4e^{2x}-3\cos x$
C. $e^{2x}+1+3\sin x$ D. $e^{2x}+1-3\sin x$
-
43. nth derivative of $f(x)=e^{3x+1}$ is: (1)
- A. $3^{n-1}e^{3x+1}$ B. $3^{n+1}e^{3x+1}$
C. $3^n e^{3x+1}$ D. e^{3x+1}
-
44. Solution set of $\sin 2x - \sin x = 0$ in $[0,2\pi]$? (1)
- A. $\{0, \pi/3, \pi, 5\pi/3, 2\pi\}$ B. $\{0, \pi/3, \pi, 5\pi/3\}$
C. $\{0, \pi/3, 5\pi/3, 2\pi\}$ D. $\{0, \pi/3, 5\pi/3\}$
-
45. If $\sin(\alpha+\theta)=12/13$ and $\sin \theta=4/5$, then $\sin \alpha = ?$ (1)
- A. $16/65$ B. $1/5$
C. $56/45$ D. $16/45$
-
46. Which is true about $f(x)=-3\cos(x/2)$? (1)
- A. Amplitude is -3 B. Range is $[-1,1]$
C. 5π is a zero D. $f(-x) = -f(x)$
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47. Angle of elevation 45° , distance 25m from base. Height? (1)
- A. 25 m B. $50/\sqrt{2}$ m
C. 50 m D. $25/\sqrt{2}$ m
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48. Which is NOT a correct identity? (1)
- A. $\sin x \cos^2 x - \sin x = -\sin^3 x$ B. $(\cos x \tan x - \sin x)/\cot x = 0$
C. $\sin^4 x - \cos^4 x = \sin^2 x - \cos^2 x$ D. $\sin x/(1+\cos x) = \sec x - \tan x$
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49. If $u=-i+2j$, $v=-3i+j$, $w=-3j$, find cosine of angle between $u+w$ and v . (1)
- A. $1/\sqrt{5}$ B. $1/2$
C. $4/\sqrt{20}$ D. 0
-
50. Given $a=i-j$, $b=2i-3j$, find unit vector in direction of $a-b$. (1)
- A. $(1/5)i - (2/5)j$ B. $-(1/\sqrt{5})i + (2/\sqrt{5})j$
C. $(1/\sqrt{5})i - (2/\sqrt{5})j$ D. $-(1/5)i + (2/5)j$
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51. Let $a = -2i+2j+mj$ and $b=3i+6j$. For what m are they perpendicular? (1)
- A. 1 B. 2
C. 0 D. -1

— END OF EXAMINATION —